PP-M-002204(Army-GL)
August 3, 1984

INTERIM FEDERAL SPECIFICATION

MACARONI WITH BEEF AND SAUCE, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers macaroni with beef and sauce thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX, Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit; Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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U.S. Department of Agriculture (USDA)

U.S. Standards for Grades of Tomato Paste

(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Meat and Poultry Inspection Regulations

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Dehydrated Onion and Garlic Association (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney Street, #410, San Francisco, CA 94108.)

Association of Official Analytical Chemists (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

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Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WN550, Environmental Protection Agency, Washington, DC 20460.)

National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.3.)

3.2 Ingredients. All ingredients shall be clean, sound, wholesome and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters, and any of the following recognizable primal and sub-primal cuts: square-cut chucks, shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, top rounds, knuckles, and bottom rounds. Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All suitable lean meat shall be used except tenderloins, and any part of a rough cut may be excluded at the option of the contractor. The beef shall be in the fresh-chilled state and shall be in excellent condition; i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class, grade and cut of meat, and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to

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dehydration, aging, or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, the meat shall show no evidence of freezing, defrosting or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the same day as boning and trimming shall be held in the temperature range of 28° to 50°F (inclusive) at all times prior to processing.

b. Holding in the fresh chilled state for not more than 4 days is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive) at all times prior to processing.

c. Holding raw beef in the frozen state is permitted provided that the beef is:

-placed in the freezer within 4 hours after certification and frozen to 0°F within 72 hours
-stored at 0°F or lower at all times
-held for not more than 60 days
-protected from freezer deterioration and damage
-stored in containers that are adequate to maintain product excellence
-maintained without defrosting until time of processing into the finished product.

3.2.2 Macaroni, elbow. The elbow macaroni shall contain glycerol monosterate and shall be made with enriched durum semolina wheat flour. The outer curvature length shall range from 0.75 to 1.0 inch, the diameter shall

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range from 0.16 to 0.20 inch, and the wall thickness shall range from 0.035 to 0.040 inch. The moisture content of the macaroni shall not exceed 13.0 percent.

3.2.3 Crushed tomatoes, canned. The canned crushed tomatoes shall be high quality, peeled, crushed tomatoes packed in puree.

3.2.4 Tomato paste. Tomato paste shall be U.S. Grade A paste conforming to the U.S. Standards for Grades of Tomato Paste.

3.2.5 Onions, diced, dehydrated. The dehydrated, diced onions shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.6 Celery, sliced, dehydrated. The dehydrated, sliced celery shall be derived from clean, sound Pascal type celery and shall be free flowing with a uniform bright color. The slices shall be 1/8 to 3/8 inch crosscut stalks with no leaf cuts. The dehydrated celery shall have a moisture content of not more than 4.5 percent and a sulfur dioxide content of 750 $\pm$ 250 ppm.

3.2.7 Salt. Salt shall be noniodized, white, refined, sodium chloride with or without anti-caking agents.

3.2.8 Starch, waxy maize, modified. The modified starch shall be white, odorless, finely pulverized, modified waxy maize food starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F, and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.9 Sugar. Sugar shall be white, refined, sucrose, granulated, cane or beet sugar, or a combination thereof. The sugar shall comply with MIL-STD-900.

3.2.10 Peppers, green, dehydrated. The dehydrated green peppers shall possess a uniform bright green color, be 1/4-inch dices made from mature sweet green or sweet-bell peppers and be free from damage caused by hail, sunburn, disease, insects or mechanical damage.

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3.2.11 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.12 Chili powder. Chili powder shall be a blend of 75-82 percent chili pepper, not less than 8 percent cumin, not less than 4 percent oregano, not less than 1 percent garlic powder, and not more than 8 percent salt. Anticaking agents may be included at a level not to exceed 2 percent. The moisture content of the chili powder shall not exceed 13 percent. The Scoville Pungency Value shall be not less than 900. The chili powder shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.13 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Powder.

3.2.14 Black pepper, ground. Ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. and shall have a characteristic penetrating odor, a hot-biting-pungent flavor and a light grey to speckled black-grey color. The volatile oil content shall be not less than 2.0 mL/100g. and the black pepper shall be of such size that not less than 95 percent passes through a U.S. Standard No. 25 sieve.

3.2.15 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindl and shall have a fragrant, clovelike aroma, strongly aromatic, pungent, clovelike flavor and a dark reddish brown color. The volatile oil content shall be not less than 3.0 mL/100g and the ground allspice shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 25 sieve.

3.2.16 Bayleaf, ground. Ground bayleaf shall be derived from the dried leaves of Laurus nobilis L. It shall possess a pleasant aromatic odor and pungent mildly bitter flavor with a pale green to yellow green color. Volatile oil content shall be not less than 1.0 mL/100g and the ground bayleaf shall be of such size that not less than 95 percent shall pass through a U.S. Standard No.30 sieve.

3.2.17 Water. Water used for ice making , formulation, and washing shall conform to the National Interim Drinking Water Regulations.

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3.2.18 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formulation.

3.3 Preparation and processing.

3.3.1 Beef preparation. The boned and trimmed beef cuts shall be coarse ground initially through a plate with holes measuring 1 inch in diameter followed by grinding through a plate with holes measuring 3/8 inch in diameter or chopped to give similar results. Grinding shall be a continuous process. The ground beef shall be blanched (see 6.4), and rendered broth and fat shall be saved for inclusion in the sauce. Unless filled immediately after blanching, the blanched beef shall be cooled to a temperature of 28° to 40°F and held within this temperature range for not more than 48 hours prior to filling.

3.3.2 Macaroni preparation. The macaroni shall be blanched in hot water until it doubles in weight and washed with cold water immediately after blanching.

3.3.3 Sauce preparation. The sauce shall be prepared from the following ingredients:

Ingredients	Percent
Beef stock and water	63.74
Crushed tomato (8 percent natural tomato soluble solids) 1/	22.00
Tomato paste (26 percent natural tomato soluble solids) 1/	4.25
Onion, diced, dehydrated	4.00
Celery, diced, dehydrated	2.00
Salt	1.25
Starch	1.25
Sugar	0.50
Green pepper, diced, dehydrated	0.40

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Ingredient	Percent
Lecithin	0.25
Chili powder	0.20
Garlic powder	0.06
Black pepper, ground	0.05
Allspice, ground	0.03
Bayleaf, ground	0.02

1/ If the specific tomato solids are not available, the tomato solids that are used must be calculated to bring the solid level to that specified. Water amount shall be adjusted accordingly.

The starch and part of the water shall be mixed to form a slurry. The remaining ingredients shall be mixed together and heated to approximately 140°F. The starch slurry shall be added, the complete mixture heated to 180°F and held for approximately 5 minutes at this temperature. The volume of the final mixture shall be adjusted with water before tray filling to compensate for evaporation loss during heating. The sauce shall be filled into the tray pack cans within 4 hours after the heating at 180°F is completed. The temperature of the sauce shall not be allowed to fall below 150°F at any time prior to filling into tray pack cans.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with blanched beef (20 percent), blanched macaroni (17 percent), and prepared sauce (63 percent) such as to conform to the finished product requirements and to the following requirements.

a. The ingredients shall be thoroughly mixed with care to provide uniform distribution without breakage of the individual ingredient pieces.

b. The temperature of the beef at time of filling shall not exceed 40°F, unless beef is filled immediately after cooking.

c. Each can shall be hermetically sealed with not less than 10 inches (Hg) vacuum immediately after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at a temperature of 240°F or below until a sterilization value (F_o) of not less than 6 has been achieved.

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3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as, but not limited to dirt, insect, insect parts, hair, wood, paper, paint, glass or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to burnt, scorched, stale, sour, rancid, musty or moldy.

c. There shall be no color foreign to the product.

d. The texture of the macaroni shall not be mushy, hard, fibrous, or tough.

e. No individual can shall contain less than 37 ounces or more than 41 ounces drained weight of meat and other ingredients except macaroni.

f. No individual can shall contain less than 38 ounces or more than 43 ounces drained weight of macaroni.

g. Sauce consistency shall be smooth without lumps and shall not be excessively thick or thin.

h. Texture of beef shall not be dry, rubbery or mushy.

i. The average net weight shall be not less than 106 ounces.

j. No individual can shall contain less than 104 ounces of product.

k. The average fat content shall not be greater than 5.0 percent.

l. The fat content of any individual can shall not be greater than 7.0 percent.

m. The salt content of any individual can shall not be greater than 1.2 percent.

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

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3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting it to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

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4.4.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or presence of any off-condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10000	50	5	6	10	11
10001 to 35000	80	7	8	14	15
Reinspection lots of 10001 to 35000	125	10	11	21	22

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TABLE II. Boned and trimmed beef defects 1/ 2/

Category		Defect
Major	Minor	
151		Presence of lymph gland measuring 1.0 inch or more in any dimension.
	201	Presence of lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension.
152		Presence of blood clot measuring 1.0 inch or more in any dimension.
	202	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.
153		Presence of bruise measuring 1.0 inch or more in any dimension.
	203	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
154		Presence of bone measuring 0.3 inch or more in any dimension.
155		Presence of cartilage measuring 0.5 inch or more in any dimension.
156		Presence of backstrap or other ligament measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	204	Presence of backstrap or other ligament measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).

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TABLE II. Boned and trimmed beef defects 1/ 2/

Category		Defect
Major	Minor	
157		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.
	205	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.
	206	Presence of heavy connective tissue on lower edge of short plate or flank.
	207	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	208	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3 square inches or more.
158		Presence of knuckle cover.
159		Presence of kidney, pizzle eye, prepubic tendon, thymus gland or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of kidney, pizzle eye, prepubic tendon, hanging tender or thymus gland measuring less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).

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TABLE II. Boned and trimmed beef defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
	210	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	211	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	212	Presence of dehydrated surface measuring 1.0 square inch or more.
	213	Presence of discolored meat (including blood disolored neck meat) measuring 1.0 square inch or more.
	214	Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension.
	215	Presence of cod, udder, kidney, or pelvic fat.

1/ Acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insect parts, hair, wood, metal) shall be determined by a Meat and Poultry Inspection Operations employee. The presence of any product not in excellent condition shall cause rejection of the involved lot.

2/ Evidence of freezing or defrosting or presence of any off-condition (including but not restricted to being off-odor, slightly sticky, gassy, rancid, or sour, or showing discoloration) shall cause rejection of the involved lot.

4.4.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine con-

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formance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.6.2 Product inspection. Eight tray packs shall be selected at random from the lot. The eight tray packs shall be heated for 15 minutes in 140° to 150°F water, opened, and inspected in accordance with table III.

a. If no defects are found, the lot shall be accepted.
b. If two or more defects are found, the lot shall be rejected.
c. If only one defect is found, a second sample of eight tray packs shall be selected at random from the lot heated as above, opened, and inspected for the defects indicated in table III. If any defect is found in the second sample, the lot shall be rejected.

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TABLE III. Product inspection 1/ 2/

Category	Defect
Major	
151	Drained weight of meat and other ingredients, except macaroni, less than 37 ounces or more than 41 ounces per tray pack can. 3/
152	Drained weight of macaroni less than 38 ounces or more than 43 ounces per tray pack can. 3/
153	Macaroni texture hard, mushy, fibrous, or tough.
154	Beef texture dry, rubbery, or mushy.
155	Sauce consistency lumpy or excessively thick or thin.

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat without breaking the meat or macaroni. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve without breaking the meat or macaroni. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a

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45° angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. After gross weight has been determined remove the macaroni and reweigh to determine weight of meat and other ingredients. The drained weights shall be reported to the nearest 0.1 ounce.

4.6 3 Fat and salt content testing. Nine tray pack cans shall be selected from the lot and distributed as follows:

- Three for laboratory analysis.
- Three for submission to the contractor.
- Three for retention by an AMS agent as a reserve sample.

The three tray packs for laboratory analyses shall individually be tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, section: Meat, except that preparation of the sample shall be as follows. The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if:

a. The average fat content of the three tray packs is greater than 5.0 percent.
b. The fat content of any individual tray pack is greater than 7.0 percent.
c. The salt content of any individual tray pack is greater than 1.2 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

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4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with nonmetallic strapping or tape is not used; (4) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (5) Length or width of pads less than box dimensions by more than 1/8 inch. (6) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.4, 3.5, and 3.6. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

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5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be not less than 1/8 inch of the full length and width dimension of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

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TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently for 35 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. Caution: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F. WARNING: Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 8 servings of 1-1/2 cup each.

5.5 *Marking*.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for approved preproduction samples (see 3.6 and 6.3).

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d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should include specific instructions in all procurement documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Blanching process. Blanching of the beef (see 3.3.1) to approximately a 70 to 75 percent yield has been found adequate in complying with finished product requirements.

Preparing activity:

Army - GL

Project No. 8940-A444

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
PP–M–002204(Army–GL)	Macaroni with Beef and Sauce, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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